

Feature Normalization Playbook

Τελευταία ενημέρωση: 2026-06-27

Αυτό το αρχείο είναι πρακτικό research reference για όλα τα canonical feature steps του `src/features/registry.py` και για τους διαθέσιμους transform / normalization helpers του `src/features/helpers/registry.py`.

Στόχος: να μη δίνουμε στο μοντέλο raw τιμές που δεν συγκρίνονται μεταξύ assets ή regimes, αλλά πληροφορία σε οικονομικά σταθερές μονάδες: returns, ATR units, percent ranks, rolling z-scores, robust z-scores, ratios, range positions, volatility-scaled values και causal lags.

Βασική αρχή

To target μπορεί να χρειάζεται raw OHLC prices. Για παράδειγμα το `directional_triple_barrier` πρέπει να ξέρει αν το `high` ή το `low` χτύπησε profit/stop levels. Αυτό δεν σημαίνει ότι το model πρέπει να δει raw `open/high/low/close`.

Η σωστή διάκριση είναι:

OHLC + ATR + side signal
-> target builder / backtest realism

normalized feature state at t
-> model input

Για meta-labeling, το πιο υγιές pattern είναι:

label = did the candidate trade reach profit before stop?

features = scale-free state when the candidate appeared

Προτεραιότητα normalizations

Priority 1: πρέπει σχεδόν πάντα να υπάρχουν

- 1. `returns`: μετατρέπει price levels σε απλές ή log αποδόσεις.
- 2. `atr_scaled_distance`: μετατρέπει price distances σε R/ATR units.
- 3. `volatility`: παράγει `atr / close` και ATR percentile.
- 4. `volatility_scaled_return`: return ανά μονάδα volatility.
- 5. `range_position`: θέση close μέσα σε rolling high/low range.
- 6. `lag`: κάνει explicit την ιστορική πληροφορία και μειώνει ambiguity για decision timing.
- 7. `rolling_zscore` με shifted stats: κάνει level-dependent columns συγκρίσιμα με την πρόσφατη ιστορία τους.
- 8. `rolling_percent_rank` με shifted window: robust regime position σε `[0, 1]`.

Priority 2: πολύ χρήσιμα για noisy intraday data

- 1. `robust_zscore`: median/MAD normalization για fat tails.
- 2. `rolling_clip`: causal winsorization για extreme spikes.
- 3. `volume_relative`: relative volume και optional volume z-score.
- 4. `rms`: energy/oscillation strength σε rolling window.
- 5. `rolling_linear_regression`: slope και R2 σε trailing windows.
- 6. `rolling_beta_residual`: idiosyncratic return vs benchmark.
- 7. `difference / slope`: momentum of indicator, όχι μόνο indicator level.

Priority 3: regime flags και model interaction helpers

- 1. `threshold_flag`: μετρά αν ένα normalized feature πέρασε meaningful level.
- 2. `between_flag`: κρατά usable range, π.χ. cycle period within bounds.
- 3. `crossing_flag`: event detection σε oscillators/filters.
- 4. `rising_flag`: direction of feature change.
- 5. `ratio`: βασικό εργαλείο για price/volatility/indicator scaling.
- 6. `reciprocal`: χρήσιμο για inverse volatility / inverse range features.

Helper catalog

Helper	Τι υπολογίζει	Πληροφορία	Γιατί είναι σημαντικό	Leakage note
<code>returns</code>	<code>close_t / close_{t-w}</code> - 1 και optional log return	cumulative move	Κάνει prices συγκρίσιμα μεταξύ assets	safe, trailing
<code>atr_distances</code>	distances από ATR-derived levels	απόσταση σε absolute/ATR context	Καλό για stop/profit geometry	safe αν ATR trailing

Helper	Τι υπολογίζει	Πληροφορία	Γιατί είναι σημαντικό	Leakage note
atr_scaled_distance	$(source - reference) / ATR$	distance σε risk units	Κεντρικό για multi-asset models	safe αν columns point-in-time
range_position	$(value - rolling_low) / (rolling_high - rolling_low)$	θέση στο πρόσφατο range	Μετρά overextension/mean reversion	rolling window trailing
realized_vol_percentile	percentile realized vol	volatility regime	Καλύτερο από raw vol level	window πρέπει να είναι trailing
robust_zscore	$(x - rolling\ median) / MAD$	robust standardized surprise	Αντέχει spikes και fat tails	default shifted stats
rolling_beta_residual	asset return minus rolling alpha/beta * benchmark	idiosyncratic move	Διαχωρίζει market beta από alpha	default shifted stats
rolling_percent_rank	rank current value vs trailing history	regime position $[0,1]$	Non-parametric, robust σε outliers	default excludes current from history
rolling_zscores	rolling mean/std z-score για πολλά columns	standardized deviation	Γρήγορο normalization πολλών features	default shifted stats
volatility	$ATR / close$, ATR percentile	relative volatility	Κάνει ATR comparable across prices	percentile currently rolling; prefer shifted alternative for strict PIT if needed
volatility_scaled_return	$return / volatility$	risk-adjusted return	Sharpe-like local signal	safe αν vol trailing
volume_relative	$volume / rolling_mean(volume)$ και optional z-score	abnormal participation	Volume scale differs across assets/sessions	default shifted stats
difference	$x_t - x_{\{t-k\}}$	change/slope proxy	Πιο χρήσιμο από raw level για filters	safe
lag	$x_{\{t-k\}}$	explicit memory	Αποφεύγει ambiguity σε current-bar features	safe
ratio	numerator / denominator	scale-free σχέση	Core normalization primitive	safe if inputs safe
reciprocal	$1 / x$	inverse intensity	Χρήσιμο για inverse vol / inverse range	handle zero/eps
rolling_clip	clip με trailing quantiles	outlier control	Σταθεροποιεί tree splits και scalars	default shifted bounds
rolling_linear_regression	rolling slope/intercept/R2	trend direction + quality	Διαχωρίζει trend από noise	trailing window
rolling_mean	trailing mean	local baseline	Για ratios/deviations	optional shift
rolling_std	trailing std	local dispersion	Για volatility/z-score	optional shift
rolling_sum	trailing sum	cumulative flow/return	Momentum over horizon	optional shift
rolling_zscore	$(x - shifted\ mean) / shifted\ std$	surprise vs history	Default normalization για unbounded columns	default shifted
rms	root mean square	signal energy	Καλή ένταση oscillators/returns	trailing
slope	$x_t - x_{\{t-window\}}$ ή fitted slope depending config	direction of change	Μετρά acceleration/turning	safe
threshold_flag	binary comparison	regime/event gate	Κάνει interpretable filters	safe
rising_flag	$x_t > x_{\{t-k\}}$	monotonic direction	Απλό trend-of-feature	safe
between_flag	value in interval	valid/neutral zone	Χρήσιμο για cycles/oscillators	safe
crossing_flag	threshold crossing	transition event	Καλύτερο από static oscillator level	safe

Feature priority groups

P0: core scale-free state για σχεδόν όλα τα models

- returns
- atr
- volatility
- vol_normalized_momentum
- return_momentum

- range_position
- regime_context
- session_context
- rolling_r2_trend_quality
- trend_slope_volatility
- volatility_regime

P1: ισχυρά price/action/trend/cycle features, θέλουν normalization

- trend
- bollinger
- roc
- adx
- vwap
- support_resistance
- support_resistance_v2
- multi_timeframe
- opening_range_breakout
- swing_extrema_context
- shock_context
- mama
- fama
- instantaneous_trendline
- supersmoother
- roofing_filter
- decycler
- decycler_oscillator
- hilbert_transform

P2: oscillators/entropy/regime diagnostics

- macd
- ppo
- rsi
- stochastic
- stochastic_rsi
- mfi
- laguerre_rsi
- fisher_transform
- inverse_fisher_transform
- sinewave_indicator
- cyber_cycle
- center_of_gravity
- even_better_sinewave
- autocorrelation_periodogram
- dominant_cycle_period
- dominant_cycle_phase
- homodyne_discriminator
- hurst_exponent
- fractal_dimension
- shannon_entropy
- permutation_entropy

P3: specialized/exogenous/flow features

- volume_features
- vpin
- order_flow_imbalance
- macro_context
- hmm_regime
- indicator_pullback
- ehlers_ml_long_candidate
- price_momentum
- zscore_momentum
- volatility_of_volatility
- parkinson_volatility
- garman_klass_volatility
- yang_zhang_volatility

Πλήρης feature-by-feature λίστα

Ο πίνακας δίνει recommended normalizations με σειρά σημαντικότητας. Όταν λέει / ATR, εννοεί helper όπως ratio ή atr_scaled_distance ανάλογα με το αν το feature είναι distance από reference ή ήδη signed value.

Priority	Feature	Τι υπολογίζει	Τι πληροφορία δίνει	Γιατί είναι σημαντικό	Normalizations / helpers έχουν νόημα
P0	returns	Απλές/log αποδόσεις από close	Τοπική κίνηση τιμής	Οι τιμές δεν είναι stationary, οι returns είναι πιο συγκρίσιμες	lag, rolling_sum, volatility_scaled_ret, rolling_zscore, rolling_percent_rank, rolling_clip
P0	volatility	Rolling/EWMA std returns	Realized risk	Volatility clustering καθορίζει edge και sizing	rolling_percent_rank, robust_zscore, ratio short/long, volatility_of_volatil
P1	trend	SMA/EMA levels	Trend anchor / mean baseline	Χρήσιμο μόνο ως relative distance, όχι raw MA level	price/MA - 1, (price-MA)/ATR, rolling_zscore, crossing_flag
P1	trend_regime	Discrete SMA relationship states	Directional regime	Απλός interpretable filter	threshold_flag usually unnecessary, use as categorical/binary
P0	lags	Delayed columns	Ιστορική μνήμη	Απαραίτητο για tabular models χωρίς sequence state	Use after normalization, n before raw price use
P1	bollinger	Rolling mean/bands/width/%B	Mean reversion + vol expansion	%B και width είναι ήδη semi-normalized	%B raw ok, bb_width/clo, bb_width/ATR, rolling_percent_rank(
P2	macd	EMA fast - EMA slow, signal, hist	Momentum acceleration	Raw MACD price units δεν συγκρίνονται	macd/close, macd/ATR, hist/ATR, rolling_zscore, crossing_flag
P1	ppo	Percentage MACD	Scale-free MACD	Καλύτερο από MACD για multi-asset	rolling_zscore, rolling_percent_rank, crossing_flag
P1	roc	Price rate of change over windows	Direct momentum	Cumulative return over lookback	already return-like, add volatility_scaled_ret, rolling_zscore, lag, threshold_flag
P0	atr	Average True Range	Absolute range volatility	Core risk unit για stops/targets	atr/close, atr_percent, atr_zscore, ATR slope, short/long ratio
P1	adx	+DI, -DI, ADX	Trend strength + directional pressure	Φιλτράρει chop vs directional regimes	ADX bounded 0-100, use DI_diff = (+DI - -DI), rolling_percent_rank(
P3	volume_features	Volume z-score, volume/ATR	Participation/activity	Volume regimes επηρεάζουν breakout/reversion	volume_relative, robust_zscore, rolling_percent_rank, volume/rolling_mean
P1	vwap	Rolling volume-weighted price	Liquidity-weighted fair value	Raw VWAP είναι price level, χρήσιμο ως anchor	(close-vwap)/ATR, close/vwap-1, rolling_zscore(dista, crossing_flag
P2	mfi	RSI-like oscillator with volume	Price-volume pressure	Βοηθά exhaustion/participation context	bounded 0-100; use /100, between_flag, crossing, rolling_percent_rank
P2	rsi	Relative Strength Index	Overbought/oversold/momentum	Bounded oscillator, good for regimes	/100, rsi-50, crossing_flag(50), between_flag, rolling_percent_rank
P2	stochastic	Close position in recent high/low	Range location	Captures close pressure within window	already 0-100; use /100, % crossing_flag, between
P2	stochastic_rsi	Stochastic transform of RSI	RSI range location	More sensitive oscillator	bounded; /100, crossings, slope

Priority	Feature	Τι υπολογίζει	Τι πληροφορία δίνει	Γιατί είναι σημαντικό	Normalizations / helpers έχουν νόημα
P1	price_momentum	Price cumulative move	Momentum/reversal	Equivalent to return over horizon	prefer return_momentum; volatility_scaled_ret rolling_zscore, lag
P0	return_momentum	Rolling sum returns	Drift over horizon	Scale-free momentum	volatility_scaled_ret rolling_zscore, rolling_percent_rank,
P0	vol_normalized_momentum	Momentum divided by volatility	Risk-adjusted momentum	Comparable across regimes/assets	rolling_zscore, rolling_percent_rank, threshold_flag, lag
P0	session_context	Hour/day/session cyclic flags	Intraday seasonality	Edges vary by liquidity/news windows	usually raw/binary ok; interactions with vol/sessi
P0	regime_context	Vol ratio, abs-return z, trend ratio/state	Market regime	Strategies are regime-dependent	already normalized-ish; ac flags, percent ranks
P1	shock_context	Large return/ATR shock and active window	Shock/reversion/continuation setup	Event state after abnormal moves	shock_strength z/percei lag, active flags, side inter.
P1	support_resistance	Rolling min/max levels	Distance to recent extremes	Breakout/retest/mean-reversion context	(close-support)/ATR, (resistance-close)/AT range_position, flags ne
P1	support_resistance_v2	Confirmed pivots, ages, touch counts, breakouts	Market structure	More realistic S/R because pivots confirmed	ATR distances, age log/cli count clip, breakout flags
P3	macro_context	Lagged macro/exogenous transforms	External regime info	Useful only if availability is modeled	z-score, pct change, explic availability lag, rolling bet residual
P1	multi_timeframe	HTF returns/trend/vol/ATR/ADX aligned to base TF	Higher timeframe context	Reduces false signals against larger trend	HTF returns already scale-ATR/close, trend scores ra lags
P1	opening_range_breakout	Session range, breakout candidates/strength	Intraday breakout structure	Opening ranges are liquidity boundaries	range/ATR, breakout distance/ATR, volume rela flags
P1	swing_extrema_context	Confirmed swing highs/lows and distances	Structural pivots	Captures market structure without raw price levels	confirmed distances/ATR, near flags, range position
P2	indicator_pullback	Pullback diagnostics from indicators	Candidate setup state	Useful as interpretable candidate generator	Normalize all distances by ATR/close; use flags/score
P2	ehlers_ml_long_candidate	Ehlers-derived long candidate	Candidate/event feature	Good for meta-label candidate side	Candidate flags raw; norm supporting Ehlers column:
P1	mama	MESA adaptive moving average	Adaptive trend anchor	More responsive than fixed MA	(close-mama)/ATR, mama spread / ATR, slope, cro
P1	fama	Following adaptive moving average	Slower adaptive trend anchor	Confirms MAMA trend	(mama-fama)/ATR, slope, crossing flag
P2	dominant_cycle_period	Estimated cycle length	Market rhythm/horizon	Helps adapt windows/filters	rolling_zscore, between_flag, lag, rolling_percent_rank
P2	dominant_cycle_phase	Cycle phase angle	Position in cycle	Useful for timing/reversal	sin/cos representation pre if available; zscore not ide
P1	instantaneous_trendline	Ehlers trendline	Smooth trend estimate	Trend anchor with less lag	(close-it1)/ATR, slope, crossing flag
P2	fisher_transform	Fisher-normalized oscillator	Turning-point sensitivity	Sharpens bounded/range signals	rolling_zscore, clip, crc zero, lag/slope
P2	inverse_fisher_transform	Maps input to bounded oscillator	Bounded nonlinear signal	Stabilizes extremes	already bounded; use cros threshold, lag
P2	sinewave_indicator	Ehlers sine/lead sine phase signals	Cycle turning context	Detects phase transitions	crossing sine vs lead, phas percent rank of amplitude present

Priority	Feature	Τι υπολογίζει	Τι πληροφορία δίνει	Γιατί είναι σημαντικό	Normalizations / helpers έχουν νόημα
P2	cyber_cycle	Ehlers high-pass/cycle oscillator	Cycle component	Removes trend to isolate swings	/ATR if price-like, rolling z RMS, crossings
P1	decycler	Ehlers decycler low-pass component	Smooth trend without cycle	Raw level is price-scale	decycler/close, (close decycler)/ATR, slope/ATR zscore distance
P1	decycler_oscillator	Fast/slow decycler spread	Trend/cycle pressure	Usually signed oscillator	/ATR, /close, rolling zscore crossing zero, RMS
P2	laguerre_rsi	Laguerre-smoothed RSI	Smooth momentum oscillator	Less noisy overbought/oversold state	bounded raw ok; /100 if p crossing, slope, percent ra
P1	frama	Fractal adaptive moving average	Adaptive trend level	Adjusts smoothing to fractality	(close-frama)/ATR, slope crossing
P2	center_of_gravity	Ehlers COG oscillator	Cycle turning/phase	Timing-oriented oscillator	rolling zscore, crossing zero slope, lag
P2	even_better_sinewave	Ehlers sinewave variant	Cycle phase/turning	Improved cycle extraction	crossings, bounded scaling RMS/amplitude normalization
P2	autocorrelation_periodogram	Dominant period/power via autocorrelation	Cycle strength and period	Tells whether cycle estimate is meaningful	period percent rank/between flag, power zscore/percent
P2	homodyne_discriminator	Ehlers phase/period discriminator	Instantaneous cycle period	Adaptive cycle horizon	period bounds, percent rank slope
P0	parkinson_volatility	High-low volatility estimator	Intrabar realized range risk	Uses OHLC range, no close-to-close only	/close if absolute, percent zscore, ratio vs ATR
P0	garman_klass_volatility	OHLC volatility estimator	Intraday volatility	Uses open/high/low/close more efficiently	percent rank, zscore, ratio ATR/realized vol
P0	yang_zhang_volatility	Open-close + overnight + range volatility	More complete volatility estimate	Useful where gaps/session opens matter	percent rank, zscore, ratio
P2	hurst_exponent	Persistence/mean-reversion statistic	Trendiness vs anti-persistence	Helps choose momentum vs reversal regimes	raw bounded-ish; percent flags above/below 0.5
P2	fractal_dimension	Price path roughness	Choppiness/complexity	Separates smooth trends from noisy paths	percent rank, flags, zscore
P2	zscore_momentum	Z-scored momentum	Standardized momentum surprise	Useful if raw momentum distribution drifts	already zscored; clip, lag, threshold
P0	rolling_r2_trend_quality	Rolling regression R2 on price	Trend quality/noise ratio	High slope alone is not enough; quality matters	R2 raw [0,1], slope/ATR, slope/price, threshold flag
P0	trend_slope_volatility	Price slope scaled by volatility	Trend per unit risk	Cross-asset trend strength	already normalized if vol correlated; add zscore/flags
P1	volatility_of_volatility	Rolling std/change of volatility	Stability of risk regime	Stops/edges degrade in unstable vol	percent rank, ratio to long zscore, rising flag
P0	volatility_regime	Volatility state flags/ratios	High/low vol regime	Controls strategy applicability	raw states ok; ratios zscore/percent rank
P3	hmm_regime	Hidden Markov regime labels/probs	Learned latent regimes	Powerful but needs strict fit/split discipline	probabilities raw, labels categorical, avoid refit leakage
P1	hilbert_transform	Amplitude/phase/frequency	Cycle amplitude and timing	Useful for Ehlers-style cycle strategies	amplitude/ATR, amplitude phase sin/cos, frequency z
P1	roofing_filter	Band-pass filtered price	Detrended cycle component	Removes trend and high-frequency noise	/ATR, rolling zscore, slope crossings
P2	schaff_trend_cycle	MACD + stochastic cycle oscillator	Trend-cycle oscillator	Fast momentum timing	bounded; /100, crossing, percent rank
P1	supersmoother	Ehlers low-lag smoothing	Smooth price/trend level	Good anchor for slope/distance	(close-ss)/ATR, slope/ss/close, crossing

Priority	Feature	Τι υπολογίζει	Τι πληροφορία δίνει	Γιατί είναι σημαντικό	Normalizations / helpers έχουν νόημα
P2	shannon_entropy	Distribution entropy over window	Uncertainty/randomness	High entropy often means noisy edge	percent rank, zscore, flags high/low
P2	permutation_entropy	Ordinal pattern entropy	Complexity of ordering	Robust nonlinear regime signal	percent rank, zscore, flags
P3	vpin	Volume-synchronized probability of informed trading	Order-flow imbalance proxy	Good for toxic flow/liquidity risk	percent rank, zscore, volume relative interaction
P3	order_flow_imbalance	Bid/ask or proxy flow imbalance	Buy/sell pressure	Direct microstructure signal when real columns exist	zscore, percent rank, rollin clip

Candle geometry features που αξίζει να προστεθούν ως helpers/combinations

Αν θες το model να μάθει candle patterns χωρίς raw price scale, χρησιμοποίησε τα παρακάτω ως derived columns:

Derived feature	Formula	Πληροφορία	Priority
bar_body_atr	(close - open) / atr_14	signed candle body σε risk units	P0
bar_range_atr	(high - low) / atr_14	intrabar volatility	P0
upper_wick_atr	(high - max(open, close)) / atr_14	rejection πάνω	P1
lower_wick_atr	(min(open, close) - low) / atr_14	rejection κάτω	P1
close_pos_in_bar	(close - low) / (high - low)	close pressure μέσα στο candle	P0
gap_open_atr	(open - prev_close) / atr_14	gap/session jump	P1
body_to_range	abs(close-open) / (high-low)	conviction vs wickiness	P1
range_percent_rank	percent rank of bar_range_atr	abnormal range	P1
body_zscore	robust z-score of bar_body_atr	abnormal directional candle	P1

Σημαντικό: αυτά πρέπει να χρησιμοποιούνται μόνο αν η απόφαση γίνεται αφού έχει κλείσει το candle. Αν μπαίνεις next_open, είναι causal. Αν μπαίνεις μέσα στο ίδιο candle, είναι leakage.

Recommended feature sets για directional triple-barrier meta-labeling

Minimal, production-first set

```
feature_cols:

- close_ret
- lag_close_ret_1
- lag_close_ret_2
- atr_over_price_14
- atr_over_price_z_252
- return_mom_8_over_vol
- rolling_r2_96
- rolling_r2_slope_96_atr
- trend_slope_vol_ratio_96
- session_europe_us_overlap

```

Γιατί: λίγα, causal, scale-free, εύκολα να γίνουν audit.

Ehlers continuation set

```
feature_cols:

- ehlers_supersmoother_slope_atr
- ehlers_roofing_atr
- ehlers_roofing_slope_atr
- ehlers_hilbert_amplitude_atr
- ehlers_hilbert_amplitude_z_252
- ehlers_hilbert_phase_64
- ehlers_hilbert_frequency_64
- dominant_cycle_period
- dominant_cycle_phase
- ehlers_decyclers_over_close
- decycler_oscillator_30_60_atr
- laguerre_rsi
- fisher_transform

```

```
- autocorrelation_periodogram_10_48_power_z
- rolling_r2_96
- rolling_r2_slope_96_atr
- atr_over_price_14
- atr_over_price_z_252
```

Γιατί: κρατά cycle/trend state, αλλά μετατρέπει price-like Ehlers outputs σε ATR/close units.

Breakout set

```
feature_cols:
- orb_breakout_distance_atr
- orb_range_atr
- close_above_vwap_atr
- volume_relative_96
- volume_zscore_96
- adx_14
- plus_minus_di_diff
- mtf_1h_trend_score
- mtf_4h_trend_score
- atr_percent_rank_252
```

Γιατί: breakouts χρειάζονται range compression/expansion, participation και higher-timeframe agreement.

Mean-reversion/shock set

```
feature_cols:
- shock_ret_z_96
- shock_atr_multiple_14
- close_pos_in_bar
- upper_wick_atr
- lower_wick_atr
- rsi_14_centered
- bb_percent_b_20_2
- distance_to_support_atr
- distance_to_resistance_atr
- volatility_regime_high
```

Γιατί: mean reversion είναι κυρίως abnormal move + exhaustion + nearby structure.

Συνδυασμοί ανά οικογένεια feature

Returns + volatility

Πρώτη επιλογή:

```
close_ret
lag_close_ret_1, lag_close_ret_2, lag_close_ret_4
rolling_sum(close_ret, 4/8/16)
rolling_sum(close_ret) / rolling_vol
rolling_percent_rank(rolling_vol)
```

Τι δίνει: local drift, recent persistence/reversal και risk-adjusted momentum.

Γιατί είναι σημαντικό: το triple-barrier label σε ATR units ευνοεί features που περιγράφουν return ανά risk unit.

Trend + volatility + quality

Πρώτη επιλογή:

```
(close - EMA) / ATR
EMA_slope / ATR
rolling_regression_slope / ATR
rolling_regression_R2
ADX
```

Τι δίνει: όχι απλά αν υπάρχει trend, αλλά αν το trend είναι καθαρό και αρκετά ισχυρό σε σχέση με volatility.

Oscillator + range

Πρώτη επιλογή:


```
RSI centered at 50
stochastic K/D
close position in rolling range
Bollinger percent_b
Fisher transform crossing zero
```

Τι δίνει: overextension, exhaustion και timing.

Γιατί είναι σημαντικό: oscillator raw values μόνα τους είναι ασταθή. Η θέση στο range και τα crossings δίνουν καλύτερο event context.

Ehlers cycle stack

Πρώτη επιλογή:

```
roofing_filter / ATR
roofing_filter_slope / ATR
decycler / close
decycler_oscillator / ATR
Hilbert amplitude / ATR
cycle phase
dominant cycle period
cycle power z-score
```

Τι δίνει: trend/cycle decomposition, timing και αν το cycle είναι αρκετά ισχυρό για να αξίζει.

Structure + levels

Πρώτη επιλογή:

```
distance_to_support / ATR
distance_to_resistance / ATR
confirmed_pivot_age
touch_count clipped
breakout/retest flags
range_position
```

Τι δίνει: proximity to liquidity/reference levels.

Γιατί είναι σημαντικό: stops/profits στο triple barrier συχνά επηρεάζονται από το αν η είσοδος είναι κοντά σε structural levels.

Volume / flow

Πρώτη επιλογή:

```
volume / rolling_mean(volume)
volume z-score
VPin percent rank
OFI rolling sum z-score
MFI centered
```

Τι δίνει: participation και flow pressure.

Γιατί είναι σημαντικό: πολλά breakouts χωρίς participation αποτυγχάνουν, ενώ shock moves με extreme flow έχουν διαφορετικό continuation/reversion profile.

Regime

Πρώτη επιλογή:

```
short_vol / long_vol
volatility percentile
volatility of volatility percentile
trend state
Hurst/fractal dimension percent rank
entropy percent rank
session flags
```

Τι δίνει: πότε το edge πρέπει να είναι ενεργό.

Γιατί είναι σημαντικό: το ίδιο setup έχει διαφορετικό expected value σε quiet trend, high-vol breakout, high-entropy chop ή session overlap.

Scaler μετά τα normalizations

Για XGBoost/LightGBM:

```
preprocessing:
  scaler: none
```

ή για συγκρισιμότητα με άλλα experiments:

```
preprocessing:
  scaler: standard
```

O scaler δεν πρέπει να είναι η βασική λύση για raw price scale. Η βασική λύση είναι τα οικονομικά σωστά feature units. Αν **standard** αλλάζει δραματικά XGBoost results, έλεγξε outliers, bad columns, leakage ή μη σταθερή κατανομή.

Για logistic/linear/neural models:

```
preprocessing:
  scaler: standard
```

Για fat-tailed normalized features:

```
preprocessing:
  scaler: robust
```

αλλά προτίμησε πρώτα **rolling_clip**, **robust_zscore** και percent ranks.

YAML snippets

ATR/close και shifted z-score volatility

```
- step: atr
  params:
    high_col: high
    low_col: low
    close_col: close
    window: 14
    atr_col: atr_14
  transforms:
    ratio:
      items:
        - numerator_col: atr_14
          denominator_col: close
          output_col: atr_over_price_14
    rolling_zscore:
      items:
        - source_col: atr_over_price_14
          window: 252
          shift: 1
          output_col: atr_over_price_z_252
```

Ehlers price-like output σε ATR units

```
- step: roofing_filter
  params:
    price_col: close
    high_pass_period: 48
    low_pass_period: 10
    output_col: ehlers_roofing_48_10
  transforms:
    difference:
      source_col: ehlers_roofing_48_10
      periods: 1
      output_col: ehlers_roofing_48_10_slope
    ratio:
      items:
        - numerator_col: ehlers_roofing_48_10
          denominator_col: atr_14
```

```

    output_col: ehlers_roofing_atr
  - numerator_col: ehlers_roofing_48_10_slope
    denominator_col: atr_14
    output_col: ehlers_roofing_slope_atr
crossing_flag:
  items:
  - source_col: ehlers_roofing_48_10
    threshold: 0.0
    direction: up
    output_col: ehlers_roofing_cross_up
  - source_col: ehlers_roofing_48_10
    threshold: 0.0
    direction: down
    output_col: ehlers_roofing_cross_down

```

Return momentum ανά volatility

```

- step: return_momentum
  params:
    returns_col: close_ret
    windows: [4, 8, 16]
  normalizations:
    volatility_scaled_return:
      items:
      - return_col: close_ret_mom_8
        volatility_col: vol_rolling_96
        output_col: close_ret_mom_8_over_vol_96

```

Candle geometry σε ATR units

Αν δεν υπάρχει dedicated feature builder, αυτά μπορούν να προστεθούν ως helpers μόνο αφού υπάρχουν οι ενδιάμεσες στήλες. Αν χρειάζεται συχνά, αξίζει dedicated feature step με tests.

```

# Conceptual recipe:
# body = close - open
# range = high - low
# upper_wick = high - max(open, close)
# lower_wick = min(open, close) - low
# normalize body/range/wicks by atr_14

```

Anti-leakage checklist

- Μην βάζεις raw `close/open/high/low` στα `feature_cols` για multi-asset model, εκτός αν είναι single-asset και υπάρχει συγκεκριμένος λόγος.
- Για rolling z-score/percentile/clip, προτίμησε shifted stats/window.
- Για current candle geometry, εκτέλεση πρέπει να είναι στο επόμενο bar/open.
- Για higher timeframe features, κράτα `shift_to_last_closed: true`.
- Για confirmed pivots/extrema, χρησιμοποίησε confirmed columns, όχι raw research labels.
- Για macro/exogenous columns, δήλωσε explicit availability lag.
- Για HMM/learned regimes, το fit πρέπει να είναι μέσα σε train fold, όχι σε όλο το dataset.
- Για scalers, fit μόνο στο train fold. Μην κάνεις global scaler πριν split.

Πρακτική σειρά υλοποίησης

1. Κράτα target/backtest OHLC raw.
2. Φτιάξε P0 normalized features.
3. Πρόσθεσε μόνο όσα P1 features εξηγούν το setup.
4. Πρόσθεσε P2/P3 features μόνο αν υπάρχει hypothesis.
5. Κάνε feature importance / SHAP / fold stability ανά feature family.
6. Αφαίρεσε raw-level duplicates και highly unstable columns.
7. Σύγκρινε `scaler: none` vs `standard` για XGBoost, όχι πριν διορθώσεις normalizations.

Short recommendation για το τρέχον US100 Ehlers config

Για `us100_30m_ehlers_deacycler_continuation_xgboost_meta_v1.yaml`, η σωστή κατεύθυνση είναι:

- target: `price_col: close, open_col: open, high_col: high, low_col: low, volatility_col: atr_14`.
- features: μόνο normalized Ehlers/cycle/trend/volatility columns.
- scaler: `none` ή `standard`, αλλά όχι ως υποκατάστατο normalization.
- πρόσθεσε/κράτα ATR-scaled versions για κάθε price-like Ehlers output.
- πρόσθεσε lags/percent ranks για returns, volatility και oscillator strength.

Με αυτό το setup το model δεν μαθαίνει raw price levels. Μαθαίνει τη σχέση μεταξύ normalized market state και πιθανότητας να πετύχει το ATR-based triple-barrier trade.